TWUN13 Programming Web Applications – Re-exam 9 January 2025 – with answers

Structure: 5 code comprehension questions (output of a code, maybe with a short explanation), 32 multiple choice questions, and one optional question.
For explanations, be concise. The exam gives a maximum of 50 points.

Grade	Points
U	< 20
3	20-29
4	30-39
5	40-50

How to read this file: the green ● option is the correct answer. The grey text in parentheses is the Swedish translation.
(Så läser du filen: det gröna ●-alternativet är rätt svar. Den grå texten inom parentes är svensk översättning.)

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1 Question 1

Which of the following statements best describes the relationship between the Internet and the Web?
(Vilket påstående beskriver bäst förhållandet mellan Internet och webben?)

Select one alternative:

- ● **The Web is a subset of the Internet that uses the HTTP protocol to provide services. ✔**
- ○ The Internet is a programming language, while the Web is an online content.
- ○ The Internet is a specific website, while the Web refers to all online content.
- ○ The Web operates independently of the Internet.

Answer:
The Internet is the network itself. The Web is one service on the Internet (pages sent with HTTP). Email is another service on the Internet.

(Internet är själva nätverket. Webben är en tjänst på Internet (sidor som skickas med HTTP). Mejl är en annan tjänst på Internet.)

Totalpoäng: 1

2 Question 2

What is the primary purpose of using an application stack in web development?
(Vad är huvudsyftet med en 'application stack' i webbutveckling?)

Select one alternative:

- ● **To provide a structured environment for developing, deploying, and maintaining web applications ✔**
- ○ To increase the cost of development
- ○ To create visual designs for the application
- ○ To limit the programming languages that can be used

Answer:
A stack (for example MERN: MongoDB, Express, React, Node) is a set of tools that work well together.

(En stack (t.ex. MERN: MongoDB, Express, React, Node) är en uppsättning verktyg som fungerar bra ihop.)

Totalpoäng: 1

3 Question 3

Which of the following is NOT a component of a typical URL?
(Vilket av följande är INTE en del av en vanlig URL?)

Select one alternative:

- ○ Protocol
- ○ Domain name
- ● **File format ✔**
- ○ Query string

Answer:
A URL has: protocol (https), domain, path, query string (?a=1) and fragment (#). "File format" is not its own part.

(En URL har: protokoll (https), domän, sökväg, query string (?a=1) och fragment (#). "Filformat" är ingen egen del.)

Totalpoäng: 1

4 Question 4

JavaScript code can be added to HTML documents
(JavaScript-kod kan läggas till i HTML-dokument)

Select one alternative:

- ○ directly within some HTML element attributes.
- ○ using script tags inside the head section.
- ● **All of the above alternative. ✔**
- ○ using script tags anywhere in the document.

Answer:
All work: in attributes (onclick="..."), in <script> in the head, and in <script> anywhere in the page.

(Alla fungerar: i attribut (onclick="..."), i <script> i head, och i <script> var som helst på sidan.)

Totalpoäng: 1

5 Question 5

Which alternative about the differences between HTML and XHTML is not correct?
(Vilket påstående om skillnaderna mellan HTML och XHTML är INTE korrekt?)

Select one alternative:

- ● **HTML and XHTML both support the same tags and same features. ✔**
- ○ In XHTML, elements may be nested.
- ○ In HTML and XHTML, elements can not overlap.
- ○ In XHTML, all attributes have to have values.

Answer:
XHTML is stricter (XML rules): all tags must be closed, all attributes need values, etc. So they are NOT the same.

(XHTML är striktare (XML-regler): alla taggar måste stängas, alla attribut måste ha värden osv. De är alltså INTE likadana.)

Totalpoäng: 1

6 Question 6

_____ is an Application Programming interface used by JavaScript to interact with HTML document in which it is contained. (fill in the blank)
(_____ är ett API som JavaScript använder för att interagera med HTML-dokumentet det ligger i. (fyll i))

Answer:
DOM (Document Object Model). With the DOM, JS can find elements (document.querySelector), change text, listen to events, etc.

(DOM (Document Object Model). Med DOM kan JS hitta element (document.querySelector), ändra text, lyssna på events osv.)

Totalpoäng: 1

7 Question 7

Which alternative about DOM events is not correct?
(Vilket påstående om DOM-events är INTE korrekt?)

Select one alternative:

- ○ Bubbling is the default event propagation mechanism.
- ○ DOM communicates to JavaScript with events.
- ● **Events are processed concurrently. ✔**
- ○ The element that dispatched the event is not always the same as the element that the listener was registered on.

Answer:
Events are handled one at a time by the event loop, not at the same time. The other three are true.

(Events hanteras ett i taget av event loopen, inte samtidigt. De andra tre stämmer.)

Totalpoäng: 1

8 Question 8

Which of the following libraries is commonly used in React for implementing routing and deep linking?
(Vilket bibliotek används ofta i React för routing och deep linking?)

Select one alternative:

- ○ Axios
- ○ Redux
- ● **React Router ✔**
- ○ Bootstrap

Answer:
Axios = HTTP requests, Redux = state, Bootstrap = CSS. React Router = routing.
(Axios = HTTP-anrop, Redux = state, Bootstrap = CSS. React Router = routing.)

Totalpoäng: 1

9 Question 9

<HashRouter> stores the current location in ____?____.
(<HashRouter> sparar nuvarande plats i ____.)

Select one alternative:

- ● **the fragment (hash) portion of the current URL. ✔**
- ○ the hierarchical portion of the current URL.
- ○ the query parameters of the current URL.
- ○ the browser address bar.

Answer:
The URL looks like site.com/#/about. Everything after # is the hash (fragment). It is never sent to the server.
(URL:en ser ut som site.com/#/about. Allt efter # är hashen (fragmentet). Det skickas aldrig till servern.)

Totalpoäng: 1

10 Question 10

What is the purpose of TypeScript's any type?
(Vad är syftet med TypeScripts typ any?)

Select one alternative:

- ● **To define a variable that can hold any type of value ✔**
- ○ To define a constant that cannot be changed
- ○ To specify that a variable must be a string or a number
- ○ To allow for strict type checking

Answer:
any turns off type checking for that variable. Easy, but you lose TypeScript's protection.
(any stänger av typkontrollen för variabeln. Smidigt, men du förlorar TypeScripts skydd.)

Totalpoäng: 1

11 Question 11

What is the primary purpose of deep linking in web applications?
(Vad är huvudsyftet med deep linking i webbapplikationer?)

Select one alternative:

- ○ To improve the performance of the application
- ○ To manage user authentication
- ● **To allow users to navigate directly to specific content or pages within the application ✔**
- ○ To serve static assets like images and CSS

Answer:
A link that goes straight to, for example, /products/42 instead of only the start page.
(En länk som går direkt till t.ex. /products/42 istället för bara startsidan.)

Totalpoäng: 1

12 Question 12

When is the useEffect hook executed by default in React?
(När körs useEffect som standard i React (utan dependency-array)?)

Select one alternative:

- ○ Only on the first render
- ○ After every re-render, except when the component's state changes
- ○ Before the component renders
- ● **After every render ✔**

Answer:
No array = after every render. [] = only once. [x] = when x changes.
(Ingen array = efter varje rendering. [] = bara en gång. [x] = när x ändras.)

Totalpoäng: 1

13 Question 13

Which type of React component is responsible for rendering UI based on the input it receives via props and does not manage any internal state?
(Vilken typ av React-komponent visar UI baserat på props och har inget eget state?)

Select one alternative:

- ○ Higher-Order Component
- ● **Functional Component ✔**
- ○ Class Component
- ○ Stateful Component

Answer:
A simple functional component takes props and returns JSX (also called stateless or presentational).
(En enkel funktionskomponent tar emot props och returnerar JSX (kallas också stateless eller presentational).)

Totalpoäng: 1

14 Question 14

What alternative about Node.js is not correct?
(Vilket påstående om Node.js är INTE korrekt?)

Select one alternative:

- ● **can only be used for creating HTTP servers. ✔**
- ○ Node modules have private scope.
- ○ includes DOM-like events and an event queue.
- ○ wraps all the OS blocking calls and provide an event interface to all OS operations.

Answer:
Node can do much more than HTTP servers: command-line tools, scripts, file handling, etc.
(Node kan göra mycket mer än HTTP-servrar: kommandoradsverktyg, skript, filhantering osv.)

Totalpoäng: 1

15 Question 15

Middleware functions
(Middleware-funktioner)

Select one alternative:

- ○ can only be used to handle GET requests and cannot process any other HTTP methods.
- ● **can end the request-response cycle. ✔**
- ○ can call the previous middleware in the stack.
- ○ can not make changes to the request and response objects.

Answer:
Middleware can send the answer itself (res.send), and then the cycle ends. It calls the NEXT middleware, not the previous one, and it can change req and res.
(Middleware kan skicka svaret själv (res.send), och då slutar cykeln. Den anropar NÄSTA middleware, inte föregående, och den kan ändra req och res.)

Totalpoäng: 1

16 Question 16

Which alternative about Node modules is correct?
(Vilket påstående om Node-moduler är korrekt?)

Select one alternative:

- ○ Node modules can not declare variables that would be global in the browser.
- ● **The EventEmitter calls all listeners synchronously in the order in which they were registered. ✔**
- ○ Node modules can only be imported from the node_modules directory.
- ○ In Node.js, concurrency should be handled explicitly.

Answer:
This is written word for word in the Node documentation. Modules can also be imported from your own files (./myfile), not only node_modules.
(Det står ordagrant i Node-dokumentationen. Moduler kan också importeras från egna filer (./minfil), inte bara node_modules.)

Totalpoäng: 1

17 Question 17

In Express, how do you set a custom response header in the HTTP response?
(Hur sätter man en egen header i HTTP-svaret i Express?)

Select one alternative:

- ○ res.setHeader("Content-Type", "text/html")
- ○ res.header("Content-Type", "text/html"
- ● **res.set("Content-Type", "text/html") ✔**
- ○ res.writeHeader("Content-Type", "text/html")

Answer:
res.set is the Express way. (res.setHeader from Node also works in practice, but res.set is the answer they want. The res.header option is missing a ")".)

(res.set är Express-sättet. (res.setHeader från Node fungerar också i praktiken, men res.set är svaret de vill ha. Alternativet med res.header saknar en ")".))

Totalpoäng: 1

18 Question 18

What does res.status(404).send("Not Found") do in an Express app?
(Vad gör res.status(404).send("Not Found") i en Express-app?)

Select one alternative:

- ○ It logs the message "Not Found" to the server console
- ● **It sends a 404 status and the message "Not Found" to the client ✔**
- ○ It sends a 404 error status and does not send any content
- ○ It sends a 200 OK status with the message "Not Found"

Answer:
status() sets the code, send() sends the text. Both go to the client.

(status() sätter koden, send() skickar texten. Båda går till klienten.)

Totalpoäng: 1

19 Question 19

Which of the following status codes indicates that an HTTP request should be re-done to retrieve the originally requested resource?
(Vilken statuskod säger att requesten ska göras om för att hämta den ursprungliga resursen?)

Select one alternative:

- ○ 200 OK
- ○ 403 Forbidden
- ○ 404 Not Found
- ● **304 Moved Permanently ✔**

Answer:
It is the only redirect option (3xx = redirect). NOTE: the exam mixed it up. Really 301 = Moved Permanently and 304 = Not Modified. Still pick this one, the others are wrong.

(Det är det enda omdirigerings-alternativet (3xx = redirect). OBS: tentan har blandat ihop det. Egentligen är 301 = Moved Permanently och 304 = Not Modified. Välj ändå den, de andra är fel.)

Totalpoäng: 1

20 Question 20

In HTTP, client errors indicate that there was a problem with the client request. Which of the following HTTP status codes is not a client error?
(Vilken av följande statuskoder är INTE ett klientfel?)

Select one alternative:

- ○ 404 Not Found
- ● **500 Internal Server Error ✔**
- ○ 400 Bad Request
- ○ 403 Forbidden

Answer:
4xx = client errors. 5xx = server errors.
(4xx = klientfel. 5xx = serverfel.)

Totalpoäng: 1

21 Question 21

Consider the following HTTP communication. Which part of it is sent by the web browser to the server and which part is sent from the server back to the web browser?
(Titta på HTTP-kommunikationen. Vilken del skickas av webbläsaren till servern, och vilken del skickas från servern tillbaka?)

```
GET / HTTP/1.1
Host: example.com

HTTP/1.1 200 OK
Content-Type: text/html; charset=UTF-8
Date: Thu, 14 Sep 2023 16:07:16 GMT
Last-Modified: Thu, 17 Oct 2019 07:18:26 GMT
Content-Length: 656

<!doctype html>
<html> ... Example Domain ... </html>
```

Select one alternative:

- ○ The first 9 lines (the header), is sent by the browser to the server. Only the HTML document is sent by the server to the client.
- ● **The first three lines are sent from the browser to the server. All other lines are sent from the server to the browser. ✔**
- ○ The first three lines are sent by the server to the browser upon connection. The following 6 lines with the communication header is then sent by the browser. Finally the server replies with the HTML page.
- ○ This is an example of an invalid HTTP request.

Answer:
The request is: "GET / HTTP/1.1", "Host: example.com" and an empty line. Everything from "HTTP/1.1 200 OK" and down is the server's answer (headers + HTML).
(Requesten är: "GET / HTTP/1.1", "Host: example.com" och en tom rad. Allt från "HTTP/1.1 200 OK" och nedåt är serverns svar (headers + HTML).)

Totalpoäng: 1

22 Question 22

Consider the following HTTP session. Which of the following best describe the communication above?
(Titta på HTTP-sessionen. Vilket alternativ beskriver kommunikationen bäst?)

Select one alternative:

- ○ Only the first line is sent by the web browser to the web server. The rest of the lines are sent by the server to the browser. The server replies by indicating its own hostname (example.com) and a status code of 404 indicating that the requested resource was not found. This is accompanied by a placeholder HTML page.
- ○ Only the first line is sent by the web browser to the web server. The rest of the lines are sent by the server to the browser. The server replies by indicating its own hostname (example.com) and a status code of 404 indicating that the requested resource was not found. The server reply is invalid because it contains an HTML document.
- ● **Only the first three lines are sent by the web browser to the web server. The rest of the lines are sent by the server to the browser. The client browser requests the document located at /page.html. The server cannot find the resource and returns with a "not found" error and a page to be displayed instead. ✔**
- ○ Only the first three lines are sent by the web browser to the web server. The rest of the lines are sent by the server to the browser. The client browser requests the document located at /page.html. The reply of the server is invalid, because "404 Not Found" status codes do not allow returning of HTML payloads, only redirects.

Answer:
Same split as Question 21: three lines from the browser (GET, host, empty line). host: example.com is sent by the BROWSER, not the server. A 404 is allowed to include an HTML error page.
(Samma uppdelning som fråga 21: tre rader från webbläsaren (GET, host, tom rad). host: example.com skickas av WEBBLÄSAREN, inte servern. En 404 får gärna innehålla en HTML-felsida.)

Totalpoäng: 1

23 Question 23

What is the primary purpose of Mongoose in a Node.js application?
(Vad är huvudsyftet med Mongoose i en Node.js-app?)

Select one alternative:

- ● **To simplify interactions with MongoDB by providing a schema-based modelling framework. ✔**
- ○ To enable JavaScript developers to write client-side code for web applications.
- ○ To provide a relational database management system (RDBMS) for Node.js.
- ○ To optimise network communication between the client and server in a web application.

Answer:
Mongoose lets you define schemas and models for MongoDB in Node.
(Mongoose låter dig definiera scheman och modeller för MongoDB i Node.)

Totalpoäng: 1

24 Question 24

In Mongoose, what is a "model"?
(Vad är en "model" i Mongoose?)

Select one alternative:

- ○ A JavaScript object that holds database configuration settings.
- ● **Are constructors compiled from Schema definitions. ✔**
- ○ A client-side JavaScript object used for rendering views in web applications.
- ○ A representation of the entire MongoDB database.

Answer:
const User = mongoose.model('User', userSchema). With User you can create, find and save documents.
(const User = mongoose.model('User', userSchema). Med User kan du skapa, hitta och spara dokument.)

Totalpoäng: 1

25 Question 25

What is the primary purpose of Object-Relational Mapping (ORM) in software development?
(Vad är huvudsyftet med ORM (Object-Relational Mapping)?)

Select one alternative:

- ○ To generate user interfaces for web applications.
- ● **To map and synchronise objects in an application to database tables and records. ✔**
- ○ To provide encryption and security for data at rest.
- ○ To manage and optimise network connections in a distributed system.

Answer:
ORM lets you work with normal objects in your code instead of writing SQL by hand.
(ORM låter dig jobba med vanliga objekt i koden istället för att skriva SQL för hand.)

Totalpoäng: 1

26 Question 26

Using HTTPS instead of HTTP can be used to protect against which type of cyber attack:
(HTTPS istället för HTTP skyddar mot vilken typ av attack?)

Select one alternative:

- ○ SQL injections
- ○ Distributed Denial of Service (DDoS)
- ● **Man-in-the-middle attacks ✔**
- ○ Buffer overflows and overruns

Answer:
HTTPS encrypts the traffic, so someone in the middle cannot read or change it.
(HTTPS krypterar trafiken, så någon i mitten kan inte läsa eller ändra den.)

Totalpoäng: 1

27 Question 27

Within information security, what is a "Denial of Service" (DoS) attack?
(Vad är en "Denial of Service" (DoS)-attack?)

Select one alternative:

- ○ A type of attack specifically designed for SaaS (Software as a Service) systems.
- ○ A type of attack in which someone poses as a legitimate member of an institution.
- ● **A type of attack in which a malicious actor makes a service or system unavailable to its regular users. ✔**
- ○ A type of attack in which a company or institution denies services to its clients to obtain economic advantage

Answer:
DoS = the service becomes unavailable, usually because it is flooded with traffic.
(DoS = tjänsten blir otillgänglig, oftast för att den översvämmas med trafik.)

Totalpoäng: 1

28 Question 28

What type of information can be stored in an HTTP cookie?
(Vilken typ av information kan sparas i en HTTP-cookie?)

Select one alternative:

- ○ Sensitive user data, such as passwords and credit card numbers.
- ○ Large multimedia files, such as images and videos.
- ● **Small pieces of text data, such as user preferences or session identifiers. ✔**
- ○ Server-side code and database credentials.

Answer:
Cookies are small (about 4 KB) and should not contain passwords or card numbers.
(Cookies är små (ca 4 KB) och ska inte innehålla lösenord eller kortnummer.)

Totalpoäng: 1

29 Question 29

Which of the following statements about session state in web applications is incorrect?
(Vilket påstående om session state i webbapplikationer är FEL?)

Select one alternative:

- ○ Session state is typically maintained using cookies or URL parameters.
- ○ Session state allows the server to store user-specific data that persists across multiple requests.
- ● **Session state can be stored on the server-side or the client-side. ✔**
- ○ Session state is often used to track user authentication and maintain user-specific settings.

Answer:
The course idea: session data lives on the SERVER, the client only has a session ID (in a cookie). Honestly the question is a bit unclear, but the other three are clearly true, so this is the answer.
(Kursens tanke: sessionsdata ligger på SERVERN, klienten har bara ett sessions-ID (i en cookie). Frågan är lite otydlig, men de andra tre är helt klart sanna, så detta är svaret.)

Totalpoäng: 1

30 Question 30

What is a cookie in the context of web development?
(Vad är en cookie inom webbutveckling?)

Select one alternative:

- ○ A server-side programming language commonly used for creating web applications.
- ○ A type of web server used for hosting static web content.
- ○ A client-side scripting language for enhancing website interactivity.
- ● **A small piece of data sent from a web server and stored on the user's device to track user information. ✔**

Answer:
The server sends it, the browser saves it and sends it back with the next request.
(Servern skickar den, webbläsaren sparar den och skickar tillbaka den med nästa request.)

Totalpoäng: 1

31 Question 31

How does a CDN improve performance for end-users accessing web content?
(Hur förbättrar en CDN prestandan för användarna?)

Select one alternative:

- ○ By serving all content directly from the origin server
- ○ By reducing the number of concurrent users
- ● **By caching content at multiple geographically distributed locations ✔**
- ○ By providing database redundancy

Answer:
The content is close to the user → shorter way → faster.
(Innehållet ligger nära användaren → kortare väg → snabbare.)

Totalpoäng: 1

32 Question 32

What is the primary purpose of load testing in large-scale web applications?
(Vad är huvudsyftet med lasttestning i stora webbapplikationer?)

Select one alternative:

- ● **To assess application performance under expected user loads ✔**
- ○ To reduce development time
- ○ To enhance visual design
- ○ To improve database design

Answer:
You simulate many users to see if the app can handle the load.
(Man simulerar många användare för att se om appen klarar belastningen.)

Totalpoäng: 1

33 Question 33

What is the output of the code below?
(Vad blir utskriften från koden?)

```
let aFuncVar = function (x) {
  console.log('Func called with', x);
  return x+1;
};
function myFunc(routine) { //passed as a param
  console.log('Called with', routine.toString());
  let retVal = routine(10);
  console.log('retVal', retVal);
  return retVal;
}
myFunc(aFuncVar);
```

Select one alternative:

- ● **Called with function (x) {...} / Func called with 10 / retVal 11 ✔**
- ○ Called with function (x) {...} / retVal undefined
- ○ Called with function routine(x) {...} / retVal 11 / Func called with 10
- ○ Called with function (x) {...} / retVal 11

Answer:
It follows the code order: 1) prints the function as text (toString), 2) routine(10) prints "Func called with 10" and returns 11, 3) prints "retVal 11". The function has no name, so it shows "function (x)".

(Den följer kodens ordning: 1) skriver ut funktionen som text (toString), 2) routine(10) skriver "Func called with 10" och returnerar 11, 3) skriver "retVal 11". Funktionen har inget namn, så det står "function (x)".)

Totalpoäng: 2

34 Question 34

a) What will happen when the "Click me!" button is clicked? b) Very briefly (ca. 50 words) explain your answer to (a).

(a) Vad händer när man klickar på knappen "Click me!"? b) Förklara kort (ca 50 ord) ditt svar.)

```
<body>
  <div id="container">
    <button id="btn">Click me!</button>
  </div>
  <pre id="output"></pre>
</body>
<script>
  function handleClick(e) {
    output.textContent += `You clicked on a ${e.currentTarget.tagName} element\n`;
  }
  document.body.addEventListener("click", handleClick, { capture: true });
  const container = document.querySelector("#container");
  container.addEventListener("click", handleClick, { capture: true });
  const button = document.querySelector("#btn");
  button.addEventListener("click", handleClick);
</script>
```

Select one alternative:

- ○ The output will display only "You clicked on a BUTTON element."
- ○ The output will display "You clicked on a BUTTON element" followed by "You clicked on a DIV element" and "You clicked on a BODY element."
- ○ The output will display "You clicked on a DIV element" and "You clicked on a BODY element," but not the BUTTON element.
- ● **The output will display "You clicked on a BODY element" followed by "You clicked on a DIV element" and "You clicked on a BUTTON element." ✔**

Answer:

b) An event first goes DOWN the tree (capture phase) and then UP (bubble phase). body and div have capture: true, so they run on the way down: BODY first, then DIV. The button's own listener runs when the event reaches the button. currentTarget = the element the listener is on.

(b) Ett event går först NER genom trädet (capture-fas) och sen UPP (bubble-fas). body och div har capture: true, så de körs på vägen ner: BODY först, sen DIV. Knappens egen lyssnare körs när eventet når knappen. currentTarget = elementet som lyssnaren sitter på.)

Totalpoäng: 2

35 Question 35

a) What will happen when the "Increase Price" button is clicked? b) Very briefly (ca. 20 words) explain your answer. c) Is there any fix to the code? Write it down.
(a) Vad händer när man klickar på "Increase Price"? b) Förklara kort (ca 20 ord). c) Finns det en lösning? Skriv den.)

```
import React, { useState } from 'react';
interface Product {
  name: string;
  price: number;
}
const ProductInfo: React.FC = () => {
  const [product, setProduct] = useState({ name: "Sample Product", price: "100" });
  const handlePriceUpdate = () => {
    setProduct({ ...product, price: product.price + 1 });
  };
  return (
    <div>
      <h1>{product.name}</h1>
      <p>Price: {product.price}</p>
      <button onClick={handlePriceUpdate}>Increase Price</button>
    </div>
  );
};
export default ProductInfo;
```

Select one alternative:

- ○ The price will not change because TypeScript will throw a compile-time error due to the wrong type.
- ● **The price will display concatenated values (e.g., "1001", "10011", etc.). ✔**
- ○ The price will increase by 1 each time the button is clicked.
- ○ The code will run, but the `price` will become `NaN` because JavaScript can't add a string and a number together.

Answer:
b) price is the string "100". string + number = the values are joined as text, not added. TypeScript does not complain, because useState did not get the type Product, so it guesses price is a string, and string + number is allowed.
c) Give the state the right type and use a number:

```
const [product, setProduct] = useState<Product>({
  name: "Sample Product",
  price: 100,   // number, not "100"
});
```

(b) price är strängen "100". sträng + tal = värdena slås ihop som text, de adderas inte. TypeScript klagar inte, för useState fick inte typen Product, så den gissar att price är en sträng, och sträng + tal är tillåtet. c) Ge state rätt typ och använd ett tal.)

Totalpoäng: 6

36 Question 36

What will be the output for the following requests made to the server?
1. GET / 2. GET /user/2 3. GET /users?age=25 4. GET /users/1 5. GET /users/3 6. GET /users?age=20
(Vad blir svaret på följande requests till servern?)

```
app.get('/', (req, res) => {
  res.send('Welcome to the Home Page!');
});
app.get('/users', (req, res) => {
  const { age } = req.query;
  const users = [{ id: 1, name: 'Alice', age: 25 }, { id: 2, name: 'Bob', age: 30 }];
  const filteredUsers = age ? users.filter(user => user.age == age) : users;
  res.json(filteredUsers);
});
app.get('/users/:id', (req, res) => {
  const userId = parseInt(req.params.id, 10);
  const users = [ ...same two users... ];
  const user = users.find(u => u.id === userId);
  if (!user) {
    return res.status(404).send('User not found');
  }
  res.json(user);
});
// (+ /about route, JSON middleware, error handler and app.listen on port 3000)
```

Select one alternative:

- ○ /: Welcome to the Home Page! | /user/2: Cannot GET /user/2 | /users?age=25: [Alice] | /users/1: {Alice} | /users/3: [] | /users?age=20: Invalid Request
- ● **/: Welcome to the Home Page! | /user/2: Cannot GET /user/2 | /users?age=25: [Alice] | /users/1: {Alice} | /users/3: User not found | /users?age=20: [] ✔**
- ○ /: Welcome to the Home Page! | /user/2: 404 Not Found | /users?age=25: [Bob] | /users/1: {Alice} | /users/3: User not found | /users?age=20: Invalid request
- ○ /: Welcome to the Home Page! | /user/2: 404 Not Found | /users?age=25: [Alice] | /users/1: {Alice} | /users/3: [] | /users?age=20: []

Answer:
/user/2 (no "s"): no route exists, so Express sends its default text "Cannot GET /user/2". /users/3: no user with id 3, so the code sends 404 "User not found". ?age=20: nobody matches the filter, so you get an empty array []. No code ever sends "Invalid request".
(/user/2 (utan "s"): ingen route finns, så Express skickar sin standardtext "Cannot GET /user/2". /users/3: ingen användare med id 3, så koden skickar 404 "User not found". ?age=20: ingen matchar filtret, så man får en tom array []. Ingen kod skickar "Invalid request".)

Totalpoäng: 5

37 Question 37

What will be printed in the console when the code is executed?
(Vad skrivs ut i konsolen när koden körs?)

```
function resolveAfter2Seconds() {
  return new Promise((resolve) => {
    setTimeout(() => {
      resolve('resolved');
    }, 2000);
  });
}
async function asyncCall() {
  console.log('calling');
  const result = await resolveAfter2Seconds();
  console.log('result: ' + result);
}
const callRet = asyncCall();
console.log('call returns: ' + callRet);
```

Select one alternative:

- ○ First, it prints out "calling," waits for 2 seconds, and then prints "result: resolved," followed by printing "call returns: undefined."
- ● **First, it prints out "calling," then prints "call returns: [Object Promise]", waits for 2 seconds, and finally prints "result: resolved." ✔**
- ○ First, it prints out "calling," then prints "call returns: [Object Promise]", and then prints "resolved."
- ○ First, it prints out "calling," waits for 2 seconds, then prints "result: resolved," and finally prints "call returns: [object Promise]."

Answer:
asyncCall prints "calling" and stops at await. The code outside continues. An async function always returns a Promise right away, so "call returns: [object Promise]" is printed. After 2 seconds the function continues and prints "result: resolved".

(asyncCall skriver "calling" och stannar vid await. Koden utanför fortsätter. En async-funktion returnerar alltid en Promise direkt, så "call returns: [object Promise]" skrivs ut. Efter 2 sekunder fortsätter funktionen och skriver "result: resolved".)

Totalpoäng: 3

38 Question 38 (optional)

Briefly explain 1. What was the latest issue/problem that YOU have encountered in the project? Please be specific. 2. How did YOU solve it?

(Förklara kort: 1. Vilket var det senaste problemet DU stötte på i projektet? Var specifik. 2. Hur löste DU det?)

Answer:

Personal question. Write about something you really did in your group project. Be specific: 1) what happened, 2) why it happened, 3) exactly what you did. Example structure: "Bookings did not show after booking a time, because the list was not fetched again. I fixed it by updating the state after the POST request." Prepare your own example before the exam.

(Personlig fråga. Skriv om något du verkligen gjorde i ert grupprojekt. Var specifik: 1) vad som hände, 2) varför det hände, 3) exakt vad du gjorde. Exempel på struktur: "Bokningar visades inte efter att man bokat en tid, för att listan inte hämtades igen. Jag löste det genom att uppdatera state efter POST-anropet." Förbered ett eget exempel innan tentan.)

Totalpoäng: 2